

LSTM Adaptive UKF

This code estimates **battery State of Charge (SoC)** by combining a **physics-based battery model** with a **machine learning model**.

First, it loads discharge data from the NASA battery aging dataset, Discharge cycles from **multiple batteries** are used to construct the training set, while data from a **separate, unseen battery** are reserved exclusively for testing.

For each discharge cycle, it builds a reference SoC curve using the fraction of discharged capacity, which decreases smoothly from full to empty during the cycle.

A **Thevenin equivalent battery model** is implemented inside an **Unscented Kalman Filter (UKF)** to estimate SoC from voltage and current measurements. The UKF predicts battery behavior and corrects itself using measured voltage.

In parallel, an **LSTM neural network** analyzes short sequences of voltage, current, voltage slope, and reference SoC to detect the **knee region** of the discharge curve, where voltage drops rapidly.

The output of the LSTM and the filter's innovation consistency are used to **adapt the UKF process noise in real time**. When the knee or model mismatch is detected, the filter becomes more responsive by increasing its uncertainty.

This hybrid approach improves SoC tracking accuracy, especially during highly nonlinear regions of battery discharge, while remaining physically interpretable and data-efficient.

Knee detection SVM Model

First, discharge data are loaded from the NASA battery aging dataset. Discharge cycles from **multiple batteries** are used to construct the training set, while data from a **separate, unseen battery** are reserved exclusively for testing.

For each discharge cycle, time-series measurements of voltage and current are extracted, and physically meaningful features are computed, including the voltage slope and curvature, which characterize the rate and acceleration of voltage decay during discharge. A smoothed slope is used to reduce measurement noise. Based on these features, a pseudo ground-truth knee label is defined, identifying the knee region as the portion of the discharge curve where voltage drops rapidly at low voltage levels.

A supervised Support Vector Machine (SVM) classifier is trained on the combined data from multiple batteries to learn generalizable knee-related degradation patterns. Feature scaling and class imbalance handling are applied to ensure stable training.

Finally, the trained SVM is evaluated on the unseen battery to assess its generalization capability. The model detects knee regions directly from the extracted features, and the predicted knee points are visualized alongside the measured voltage curve, enabling clear identification of knee onset during discharge.